

# Team Worksheet — Lab 12

## Attribute MSA and Kappa Analysis (Black Belt Only)

Team members: \_\_\_\_\_

Date: \_\_\_\_\_

### 1. Our understanding of the problem

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### 2. Our working

Use the data in lab-12-attribute-msa-kappa.xlsx.

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### 3. What we produced

*Deliverable: An attribute agreement study with within-, between- and versus-standard Kappa values.*

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### 4. What the evidence does NOT tell us

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### 5. Our conclusion

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### 6. Step notes

**Step 1.** Establish why Gage R&R fails for attribute data: there is no continuous scale on which to compute variance components.

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**Step 2.** Identify a judgement-based measurement in your capstone — visual inspection, cosmetic grading, document approval, claim adjudication or defect classification.

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**Step 3.** Select 30 to 50 sample items spanning clear-pass, clear-fail and deliberately borderline cases. Borderline items are where agreement actually breaks down.

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**Step 4.** Have an expert establish the KNOWN STANDARD (true) classification for every item. Without a standard you can measure consistency but never correctness.

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**Step 5.** Have 3 appraisers each classify every item twice, in randomised order, blinded to their own previous answer and to each other.

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**Step 6.** Calculate WITHIN-appraiser agreement — does each inspector agree with themselves? Self-disagreement is the most common and most ignored failure.

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**Step 7.** Calculate BETWEEN-appraiser agreement — do the inspectors agree with each other?

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**Step 8.** Calculate agreement VERSUS THE STANDARD — are they collectively correct, or consistently wrong together?

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**Step 9.** Compute Cohen's Kappa for appraiser pairs and Fleiss' Kappa across all appraisers. Kappa corrects raw agreement for agreement expected by chance.

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**Step 10.** Apply the acceptance rule: Kappa  $\geq 0.90$  excellent, 0.75-0.90 acceptable, below 0.75 requires action. Note that 90% raw agreement can still yield a poor Kappa.

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**Step 11.** Where agreement fails, prescribe the fix: boundary samples, photographic standards, tightened operational definitions or appraiser retraining. Then re-run.

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## 7. Self-check

Your study includes borderline items and a known standard, you have computed within-appraiser, between-appraiser and versus-standard agreement plus Kappa, and you have an action plan for any Kappa below 0.75.

Complete? ■ Yes ■ Not yet

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